

Meeting 1203

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Index

1 11.26 Discussion

2 11.26 Todo

3 Result - What I do this time

11.26 Discussion

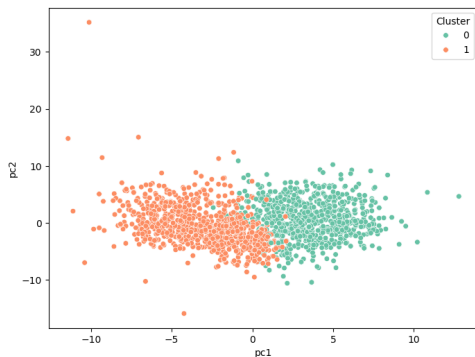
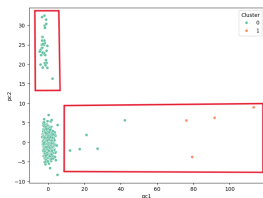
Semiconductor Manufacturing Process

- ▶ 抽掉 outlier 讓 PCA 更散更清楚
- ▶ (Y_i, X_i) iid $F(\cdot)$

11.26 Todo

- ▶ 繼續分成幾群再套模型比較最合適的
- ▶ 對 outlier 做質性的統計方法
- ▶ 在降維分群之前做 SMOTE 看看效果

Results I PCA + KMeans, 2 clusters



cluster	0	1
$y = 0$	660	770
$y = 1$	58	37

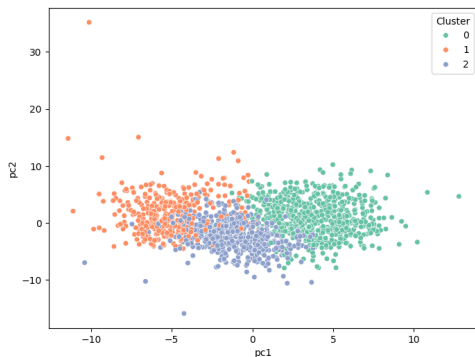
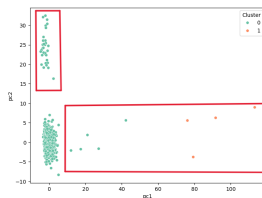
Results I 5-folds cv - mean

Summary

model	ACC	FAR	ROCAU
LR	0.8551	0.0977	0.5476
SVM	0.9292	0.0048	0.4976
DT	0.8264	0.1292	0.5366
RF	0.9317	0.0027	0.5036
XGB	0.9247	0.0130	0.5176

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8315	0.1136	0.5470	0.9021	0.0636	0.5593
SVM	0.9178	0.0015	0.4992	0.9542	0.0000	0.5000
DT	0.7967	0.1561	0.5515	0.8773	0.0896	0.5481
RF	0.9178	0.0015	0.4992	0.9554	0.0000	0.5143
XGB	0.9081	0.0182	0.5258	0.9554	0.0000	0.5143

Results II PCA + KMeans, 3 clusters



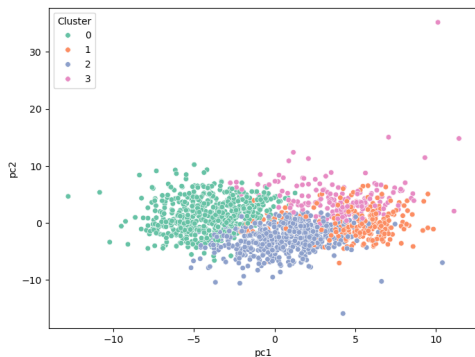
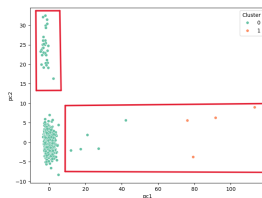
cluster	0	1	2
$y = 0$	592	327	511
$y = 1$	49	19	27

Results II 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8346	0.1082	0.5170	0.8872	0.0673	0.5247
SVM	0.9236	0.0000	0.5000	0.9422	0.0031	0.4985
DT	0.8081	0.1436	0.5415	0.8555	0.1042	0.5229
RF	0.9235	0.0017	0.5092	0.9451	0.0000	0.5000
XGB	0.9142	0.0135	0.5132	0.9335	0.0122	0.4939

model	C_2 ACC	C_2 FAR	C_2 ROCAU
LR	0.9183	0.0470	0.6032
SVM	0.9498	0.0000	0.5000
DT	0.8755	0.0900	0.5683
RF	0.9498	0.0000	0.5000
XGB	0.9405	0.0137	0.5298

Results III PCA + KMeans, 4 clusters



cluster	0	1	2	3
$y = 0$	513	231	523	163
$y = 1$	39	12	32	12

Results III

0.5test, ns=3, 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8424	0.1054	0.5277	0.9054	0.0520	0.5073
SVM	0.9294	0.0000	0.5000	0.9507	0.0000	0.5000
DT	0.8081	0.1424	0.5038	0.9054	0.0607	0.6030
RF	0.9294	0.0000	0.5000	0.9507	0.0000	0.5000
XGB	0.9149	0.0156	0.4922	0.9507	0.0000	0.5000

model	C_2 ACC	C_2 FAR	C_2 ROCAU	C_3 ACC	C_3 FAR	C_3 ROCAU
LR	0.8955	0.0650	0.5937	0.8743	0.0862	0.6236
SVM	0.9423	0.0000	0.5000	0.9314	0.0000	0.5000
DT	0.8595	0.0956	0.5094	0.8171	0.1225	0.4387
RF	0.9423	0.0000	0.5000	0.9314	0.0000	0.5000
XGB	0.9441	0.0019	0.5276	0.9257	0.0062	0.4969

The End

Questions & Comments